



# Jaume Ivars Grimalt

MACHINE LEARNING ENGINEER · COMPUTER VISION & PRODUCTION SYSTEMS

Valencia, Spain · remote or hybrid · 20 h/week for freelance and part-time work

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## Profile

Most machine learning projects stall between a working notebook and something a user can open. I cover the whole path: training the model, putting it in production, and building the interface it is used through. At Mycrospace that took high-density colony counting from a Master's thesis to a product running in **5 laboratories**. I work the same way for clients, either the full path or the part a team is missing.

## Technical Skills

|                            |                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------|
| <b>Computer Vision</b>     | Object detection, semantic segmentation, density estimation, classification                     |
| <b>LLM integration</b>     | LangGraph, LangChain, Anthropic API, N8N                                                        |
| <b>Training</b>            | Python, PyTorch, Hugging Face, timm, multi-GPU training on HPC (SLURM, checkpoint-resumed jobs) |
| <b>Production</b>          | Docker, GCP (Cloud Run, Cloud SQL, GCS), Hetzner + Dokploy, BullMQ/Redis, CI/CD                 |
| <b>Product Engineering</b> | NestJS, Prisma, PostgreSQL, Next.js, TypeScript, React Native/Expo, Stripe, PostHog             |

## Experience

### Mycrospace

Valencia, Spain

Co-FOUNDER & CTO

2024 – Present

- Computer vision for microbiology. Started in 2024 as my Master's thesis; the company was incorporated in January 2026.
- Detection ran on Ultralytics, which is AGPL-3.0 and rules it out for a closed SaaS. So we rebuilt it: the detector starts from the YOLO family of architectures and is trained from scratch on our own data, with our own modifications, on a base we can license.
- Colony detection sits at **67 mAP50-95**. Against counts from expert microbiologists, the product reaches **95% median accuracy**.
- Crowded plates are the hard case. Past roughly **300 overlapping colonies** the detector stops separating instances, so for that regime I reformulated counting as density estimation and trained density map regressors with Bayesian priors.
- 1,500** images annotated out of a **5,000**-image corpus of plates we shot ourselves, across **12** microbial species, under protocols written with microbiologists and custom weakly-supervised labelling tools. More than **3,000 GPU-hours** of training on Leonardo, under **two compute allocations** awarded through competitive EuroHPC calls in the computational biology domain.
- I built the rest too: inference workers on BullMQ and Redis, a NestJS/Prisma API over PostgreSQL, a Next.js web client, a React Native app for the bench, Stripe, PostHog. It ran on GCP until the bill outgrew the traffic, then moved to Hetzner with Dokploy. LLMs internally for automations and lead search.
- 5 laboratories** run it in production. I supervise one full-stack developer.

### MIALAB — Medical Imaging Analysis Laboratory, UPV

Valencia, Spain

ML ENGINEER, PART-TIME

May 2025 – Jan 2026

- Volumetric brain MRI segmentation with nnU-Net, comparing multi-scale feature-extraction strategies.
- VolBrain has processed **700k+** MRI volumes, so inference cost scales. Profiled the pipeline from NIfTI ingest to voxel-level prediction to find where the time went.

### Tapstar

Valencia, Spain

FREELANCE FULL-STACK ENGINEER

Jan 2025 – May 2025

- Tapstar sells QR cards and stands to restaurants, shoe shops and other high-street businesses. I built the software around them: QR generation, the back office where a client manages their products, the frontend, and the SEO.
- The scan is where the data starts: analysis of the profiles behind it, and ratings broken down per employee.

### Vesta-Z

Valencia, Spain

FREELANCE DATA ENGINEER

2023

- Full-stack feature work on their cloud product, and SEO for parts of tractors, trucks and other construction equipment.
- Custom database schema, designed and implemented, for cross-matching items across different providers.
- An LLM customer-support bot for their users.

**Neurocatching**, ML Engineer. Temporal models over raw ocular-movement sequences

Valencia, Spain · 2022 – 2023

**Sciling**, AI Consultant. Conversational systems on GPT-3-era LLMs, applied NLP, generative models

Valencia, Spain · 2022

## Projects

### ConvNeXt-V1 trained from scratch

OWN TRAINING FRAMEWORK ON EUROHPC (LEONARDO BOOSTER)

- Trained on **4 GPUs** on Leonardo Booster with a training framework I wrote, not the timm script. Leonardo cuts jobs at **12 hours** of wall clock, far short of a 300-epoch run, so the training runs as a chain of jobs resuming from checkpoint.
- ConvNeXt-Tiny, **30M** parameters, **81% top-1** on ImageNet-1k over **300 epochs** with timm augmentation. It replicates the original paper, which is the point: it gives me a baseline I own. [github.com/jaume2000/convnext\\_v1](https://github.com/jaume2000/convnext_v1)
- In progress on that baseline: finding redundant parameters, and reparametrising residual blocks interpreted as steps of an ODE solver.

# Education & Awards

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**M.Sc. in Artificial Intelligence, Computer Vision & Digital Image**

Valencia, Spain

UNIVERSITAT POLITÈCNICA DE VALÈNCIA · GPA 8.7/10 · THESIS AWARDED HONOURS, AND IT BECAME MYCROSPACE

2023 – 2024

**B.Sc. in Computer Science**

Valencia, Spain

UNIVERSITAT POLITÈCNICA DE VALÈNCIA · GPA 8.3/10 · HONOUR MENTIONS IN MACHINE LEARNING AND STATISTICS

2019 – 2023

**1st Prize, IdeasUPV Startup Competition**, Mycrospace, €2,500 grant

Valencia, Spain · 2024